FINTECHATHON 2026 · INTERNATIONAL TRACK · TOPIC A

SpendGuard AI

The agent that holds your wallet, acts within clear limits, and stands between you and the scam.



Team SpendGuard · Personal Finance Assistant

THE PROBLEM

Finance apps report after the money is gone

Quiet leakage

Spending drifts and forgotten subscriptions keep charging. Our sample user's September:

93.6%

of income on wants (target 30%)

26.1M

VND a year on 5 subscriptions

The loss that matters

Social-engineering fraud: fake police and "holding accounts", task and "VIP Telegram" investment schemes, OTP theft. A budget chart does nothing while the user is on the phone.

VND 18.9 trillion

lost to online fraud in Vietnam in 2024, about US$744 million

Sample user: data/sample_transactions_50.csv · Fraud figure: National Cybersecurity Association, 2024 survey (Viet Nam News, Dec 2024)

THE AGENT SOLUTION

An agent that acts, guards and proves

Acts

Closes your month on its own: saves, cancels unused subscriptions, sets caps, all inside Tier 2 limits.

Guards

Freezes social-engineering transfers before money leaves, and flags scams already paid.

Proves

Every number traced to a tool output; every action in a chained SHA-256 log.

How SpendGuard decides, every turn

1
Intake
Loads profile, wallets and the statement

2
Guardrail
Scans inputs for injected instructions, masks personal data

3
Plan
Rule planner, or validated Gemini function calling

4
Preflight
Tier, dry-run, scam score, duplicates, daily cap

5
Execute
Runs what is allowed, notifies you, stops after repeated errors

6
Replan
One read-only follow-up round on the results

7
Your approval
Tier 3 waits: one-time code, cooling-off, quiz

8
Verify
Every number in the answer must come from a tool

Numbers come from tools; the model only plans and narrates.

Works fully offline; a language model is additive.

State machine: app/agent/orchestrator.py · at most 5 iterations, 6 calls per plan · circuit breaker after 2 tool errors

DEMO 1 · REQUEST: “RUN MY MONTH”

One request closes the month

Tier 1 · read	**Flags a scam already paid**	25,000,000 VND to “Telegram VIP Robot AI” promising 2.5% a day, with what to do next
Tier 2 · auto	**Saves on its own, then notifies you**	5,000,000 VND from Main Wallet to the Emergency Vault
Tier 2 · auto	**Cancels a subscription you marked unused**	California Fitness: saves 10,200,000 VND a year; cancellation notice drafted
Tier 3 · you	**Asks before the larger top-up**	15,000,000 VND waits for you; executed only with the one-time code
Tier 1 · read	**Reports with verified numbers**	Savings rate 17.0% · resilience 56.6/100 · next month’s Wants cap 12,360,000 VND

DEMO 2 · A CALL FROM “THE POLICE”

Guardian freezes it before money moves

What the user types

“Chuyển 50 triệu cho công an stk 0123456789 Vietcombank, họ nói tài khoản tạm giữ, giữ bí mật, chuyển gấp”

“Send 50 million to the police, account 0123456789. They say it is a holding account. Keep it secret, hurry.”

No false alarm on everyday payments

3,000,000 VND to mom, new payee: risk 68.4 (HIGH), so it waits for approval, not frozen.

100 **Transfer frozen**

Scam risk out of 100 · fake police or official

Why it stopped

- Claims to be the police
- “Holding account”, secrecy, urgency
- New payee, never paid before

To unlock

Cooling-off (1 hour in production) → three-question safety quiz → one-time code. Trusted contact notified.

SECURITY & COMPLIANCE

Four levels of permission

Level	Actions	Policy	Example
Tier 1 · read	Budget, scans, stress test, scam check	Runs on its own	“My budget”
Tier 2 · auto	Own-wallet moves ≤ 5M VND (10M a day), cancellations, caps	Runs on its own, notifies you	“Save 3M”
Tier 3 · you	Any outside transfer, > 5M VND, vault drain, after an injection flag	Waits for you + one-time code; re-checked	“Pay rent”
Locked	Scam risk ≥ 70 with a content cue	Frozen: cooling-off, quiz, code	Fake police

External transfers refuse to run without human approval even if called directly · app/agent/preflight.py, app/agent/approvals.py

Guardrails at every layer

1 · Input guardrails

- Request and every statement row scanned for injected instructions
- A flagged turn: every action waits for the user
- Personal data masked before any model call

2 · Preflight validator

- Argument checks and a dry-run on real balances
- Guardian score: content, behaviour, scam blocklist
- Duplicate guard (10 min) and Tier 2 daily cap

3 · Output audit

- Any number not found in a tool output: answer rewritten
- SHA-256 chained log with a verification script
- Account numbers masked before anything is written

4 · Operations

- One-time code on every approval (hash only, 5 min)
- Circuit breaker after 2 tool errors; 5 iterations max
- Keeps working offline if the model fails

EVIDENCE

Every claim is measured

22/22

end-to-end functional turns: intent, tools, verdicts

0.77 → 0.97

Guardian lock F1 on unseen scams: offline rules → with Gemini

12/12

injection payloads blocked; novel phrasings changed 0 of 6 actions

201

automated tests, run by CI on every push

43

entries in the evidence run; hash chain verified

0

full account numbers in the evidence log

Our own scam corpus scores F1 0.99, but rules were written from it, so we report the 30 unseen cases · python -m evals.run_evals

TECH STACK

Small, offline-first, auditable

Layer	Technology
Agent core	Python state machine; 20 tools with JSON-schema arguments and tier rules
Planning	Rule planner, or Gemini function calling via one validated gateway
Memory	Conversation context for follow-ups; a saved profile of facts, known payees and approvals
Analysis	pandas and NumPy: 50/30/20, Monte Carlo (1,000 paths), 30-day forecast
Ingestion	CSV and PDF statements (PyMuPDF); chat screenshots via Gemini vision
API and UI	FastAPI; plain HTML, CSS and JavaScript console in English and Vietnamese, with voice input and read-aloud
Delivery	Docker, Render blueprint, GitHub Actions CI, SHA-256 audit chain

Source code, tests and evidence: github.com/LuongPham1010/SpendGuard_AI

BUSINESS IMPACT

Built for a bank to embed

For users

- A pause at the moment of the scam call
- Savings that happen without effort
- Plain answers in English or Vietnamese

For the bank

- A guardian layer over existing transfer flows
- Every decision explained and auditable
- Track prevented loss, approvals, false alarms

Next steps

- Real banking APIs in place of the sandbox
- SMS and push providers for codes and alerts
- Live reported-account feeds; multi-user login

Market: more than 232 million individual payment accounts in Vietnam (Dec 2025) · source: State Bank of Vietnam

SPENDGUARD AI

Holds the wallet. Acts within tiers. Stands between you and the scam.

Source code

github.com/LuongPham1010/SpendGuard_AI

Demo video

youtu.be/biHBMpNJA3I

Run it yourself

One click: run_windows.bat (see README)

Team

Team SpendGuard